

ECON 630

Fall 2026

TTH 7:00 PM - 9:00 PM | 08/25/26 - 10/16/26

Lecture

Hilltop Campus | Cowell Hall | Room 314

Instructor Information

Tom Bowen

tbowen@usfca.edu

Office Hours

Thursday 5:45 to 6:30 pm

Office Location: McLaren 112

Course Description

Special topics in econometrics, analytics, or empirical economic methodology not covered in other course

Course Learning Outcomes

During the course, students will improve their programming skills, learn how to use AI effectively to program, and perform optimization exercises using the maximum likelihood method. Students will learn how to translate theoretical economic models into estimable econometric models.

Assignments

Homework assignments are designed to improve your understanding of concepts. No late homework will be accepted. You may discuss homework assignments with your classmates, but you should write up the assignment individually. Homework assignments will be posted on Canvas one week prior to the due date.

Grading Breakdown and Grading Policies

Attendance and Participation	20%
Problem Sets	20%

Midterm Exam	40%
In-Class Quiz (every other Thursday)	20%

Final Grades will be assigned according to the following table, if you are on the threshold, you will get the higher grade (i.e. if you get exactly 93% you will get an A, not an A-).

Grade	Score
A	93%+
A-	90%-93%
B+	87%-90%
B	83%-87%
B-	80%-83%
C+	77%-80%
C	70%-77%
D	60%-70%
F	<60%

Texts and Supplies

Introduction to Python for Econometrics, Statistics and Data Analysis

Authors: Kevin Sheppard

Where to purchase (URL):

https://bashtage.github.io/kevinsheppard.com/files/teaching/python/notes/python_introduction_2021.pdf

Course Schedule

Week	Activity
Week 1	Introduction
Week 2	Intro to python: list, loops, functions, logical tests,
Week 3	Numpy and Pandas
Week 4	Data Visualization in Python
Week 5	Claude Code and AI for economic programing

Week	Activity
Week 6	Solving simple economic models
Week 7	Solving simple economic models
Week 8	Review and Midterm Exam

Program Learning Outcomes

Economic Data Manipulation: Students will be able to effectively use modern programming languages to clean, organize, query, summarize, visualize, and model large volumes and varieties of data.

Use AI coding tools effectively for economics: generating and debugging code with AI assistance while retaining the judgment to verify that the output is correct, and the economic understanding to know what "correct" means.

Economic Data Analytics: Students will possess a theoretical and applied understanding of the use of econometrics and statistics for descriptive and causal inference.

Applied Economic Theory: Students will be able to understand and apply economic theory to understand how businesses and other organizations interact with each other and with users/customers/clients and use this understanding to guide data analysis.

Economic Problem Solving: Students will be able to solve real-world data driven business and policy problems working with economists, policy makers, data scientists and business practitioners.

Economic Communication: Students will be able to communicate their research approach and findings at an excellent level, both in writing and verbally.

University Policies

Credit-hour Policy

One unit of credit in lecture, seminar, and discussion work approximates one hour of direct faculty instruction (or 50 minutes plus a break) and a minimum of two hours of out-of-class student work per week through one 15-week semester. For further details, see USF's [Credit Hour Policy](#).

The below resources and additional information can be found in the [Student Life Resource Toolkit](#).

Students with Disabilities

The University of San Francisco is committed to providing equal access to students with disabilities. If you are a student with a disability, or if you think you may have a disability, please contact Student Disability Services (SDS) at sds@usfca.edu or (415) 422-2613, to speak with a disability specialist (please note all communication with SDS is private and confidential).

If you are eligible for accommodations, please request that your accommodation letter be sent to me as soon as possible (students are encouraged to contact SDS at the beginning of the semester), as accommodations are not retroactive. Once I have been notified by SDS of your accommodations we can discuss your accommodations and ensure your access to this class or clinical setting. For more information please visit the [SDS website](#).

Behavioral Expectations

The University of San Francisco is committed to providing an environment consistent with the academic nature and core values of the institution in which students can participate in learning as a humanizing, social activity rather than a competitive exercise to support the full, integral development of each person and all persons, with the belief that no individual or group may rightfully prosper at the expense of others.

It is important you know what is expected of you academically and behaviorally through the published course syllabus, the [Student Conduct Code](#), and other instructions provided by the instructor. Therefore, you are expected to uphold the following:

- Behave in accordance with the Student Conduct Code and other USF policies
- Refrain from disrupting the ability of fellow students to learn or the instructor's ability to teach.
Examples of disruption include:
 - Cell phone or computer use that significantly or repeatedly distracts others
 - Coming to class late, leaving early, or excessively physically relocating oneself in the classroom
 - Speaking frequently without being called on
 - Yelling, cursing, or engaging in aggressive behavior
- When interacting online, communicate in a respectful fashion. This includes, but is not limited to:
 - Refraining from name calling, profanity, or typing in all capital letters
 - Sending multiple emails with one sentence
 - Avoiding rants or discussing non-relevant topics

Open discussion and disagreement are encouraged when done respectfully and in the spirit of academic discourse. There are also a variety of behaviors that, while not against a specific University policy, may create disruption in this course. Students whose behavior is disruptive or who fail to comply with the instructor may be dismissed from the class for the remainder of the class period and may need to meet with the instructor or Dean prior to returning to the next class period. If necessary, referrals may also be made to the Student Conduct process for violations of the Student Conduct Code.

Academic Integrity

As a Jesuit institution committed to *cura personalis*—the care and education of the whole person — USF has an obligation to embody and foster the values of honesty and integrity. All members of the USF

academic community are responsible for maintaining the standards of honesty and integrity. The [honor code](#) applies to all students (undergraduate and graduate) in the College of Arts and Sciences, the School of Education, the School of Management, and the School of Nursing and Health Professions. Faculty and students in the School of Law should review their own [honor code](#) for policies and procedures. Students enrolled in distance learning (online courses) are subject to these policies as well as supplemental policies set forth by their program. All students should review and familiarize themselves with the honor code, prohibited conduct, and procedures.

Counseling and Psychological Services

Many college students experience mental health difficulties. Counseling and Psychological Services (CAPS) is a great source of support for addressing a range of mental health needs, including anxiety, stress, loneliness, relationship struggles, grief/loss, adjusting to change, identity development, racial/cultural concerns, and mild to moderate depression.

CAPS offers in-person and remote individual and group therapy to enrolled students. Students seeking services are scheduled for a 15-20-minute phone screening to assess immediate risk, as well as identify initial treatment needs, and make recommendations for next steps. These may include a crisis or risk assessment, brief individual therapy (every 2-3 weeks), single session therapy, weekly group therapy, or referrals to off-campus providers. Please note, CAPS does not have a psychiatrist on staff to prescribe medication. Students in need of medication or with more complex mental health concerns are referred off-campus for specialized, comprehensive, treatment and/or ongoing, longer-term therapy. To make an appointment, call CAPS at 415.422.6352 or request an appointment via the CAPS website. CAPS does not accept walk-in appointments.

Students with urgent mental health needs can call the CAPS All-Hours line (855-531-0761) to immediately speak with a mental health clinician. The CAPS All Hours line is available 24/7, including weekends/holidays, and accepts international calls, and should only be accessed for mental health crises. For all other mental health concerns, including scheduling, canceling, rescheduling, or leaving a message for a CAPS clinician, please call CAPS at 415.422.6352 during normal business hours.

All students are encouraged to check out the [CAPS website](#) and access our extensive online resources, podcasts, mental health apps, videos, self-care strategies, and more.

If you are concerned about a student and would like someone to follow up, please contact the Dean of Students Office at 415.422.5330. If you know someone who is an immediate risk of harming themselves or others, please contact Public Safety at 415.422.2911 in San Francisco, out of state or additional locations within California dial 911, or call the National Suicide & Crisis Lifeline by dialing 988.

Title IX

The Title IX Office seeks to stop, remedy, and prevent occurrences of sex and gender-based discrimination, sexual harassment, and sexual violence. The University has a [Policy](#) on Nondiscrimination Based on Sex and Gender, Sexual Harassment, and Sexual Misconduct for all Students, Employees, and Third-Parties. If you have experienced any behaviors related to this policy, we encourage you to report

the incident. If you report these behaviors to any staff or faculty member, they must notify the USF Title IX Coordinator.

Students who wish to report any sexual misconduct should use the online mandatory reporting [form](#). You can also contact the Title IX Office directly by calling (415)422-4563, emailing titleix@usfca.edu, or visiting our office in Lone Mountain, Room 145.

As an employee at USF, and your Professor, I am a mandatory reporter, meaning I have to share any instances of sexual harassment or sexual violence shared with me or that become known to me. I will have to share this information, including names and any details known, to the Title IX Office to connect you with resources. If you would like more information about the resources available, you can ask me at any time this semester. You do not need to tell me why you are asking to get help for a friend, another student, or yourself.

If you are interested in accessing confidential resources, please consider these options. If you report any information to the following, it will not generate a report to the Title IX office:

- Students may speak to someone confidentially by contacting Counseling and Psychological Services at (415) 422-6352, or speaking to a clergy member in University Ministry at (415) 422-4463.
- To find more about reporting sexual misconduct (including sexual assault, relationship violence, or stalking) and resources at USF, visit the USF Title IX website.
- For off-campus resources, contact San Francisco Women Against Rape (SFWAR) at (415) 647-7273; www.sfwar.org

If during the course of the semester you need accommodations related to pregnancy or parenting, please let me know and I can connect you to the Title IX office. They are able to offer support through arranging accommodations, leave, or other support. The Title IX office is responsible for ensuring pregnant and postpartum students get the changes they need to stay healthy while being able to continue their education.

Confidential Resources for Reporting Sexual Misconduct

- Students may speak to someone confidentially which will not generate a report to the Title IX Office by contacting Counseling and Psychological Services at (415) 422-6352 during M-F 9-4pm, or speaking to a clergy member in University Ministry at (415) 422-4463.
- If you need to speak to a mental health clinician immediately, please call the CAPS 24/7 All Hours Line at 855-531-0761 (available daily, including weekends and holidays, and accepts international calls), Public Safety (415-422-2911), 911, the Suicide Hotline (dial 988), or go to your nearest emergency room
- For off-campus resources and local Bay Area organizations check [here](#).

Learning, Writing, and Speaking Centers

The Learning, Writing, and Speaking Centers (LWSC) at USF provide individualized support to assist students in better understanding course material and to aid them on their path to success. Services are free and include tutoring, collaborative peer support services, academic skills coaching, writing, and speaking support. Services are available in-person and on Zoom.

LWSC staff can be reached Monday through Thursday between 9:00am - 8:00pm and Friday between 9:00am-5:00pm at LWSC@usfca.edu or through the chatbox on [our myUSF webpages](#) or by phone at (415) 422-6713. To make an appointment for subject tutoring, academic skills coaching, the writing center, or the speaking center, students should visit the [Student Appointment Dashboard](#).

Communication

All course communications, like all other USF communications, will be sent to your USF official email address. You are therefore strongly encouraged to monitor that email account.

Gleeson Library

Looking for help with a research paper or project? Set up a consultation with a Librarian or get 24/7 research help [online](#).

Attendance Policy

Attendance and participation accounts for 20% of the final grade.

Additional USF Resources

USF Food Pantry

The USF Food Pantry is an intermediate, short-term solution for any registered USF student to receive food and toiletry resources. Students are invited to stop by the pantry located on the first floor of Gleeson Library in the Atrium, and take the items that they need. Items are available on a first-come, first-serve basis until our supply is depleted. You will be asked to check-in via QR code before entering the pantry. For more information and the current schedule, visit the [USF food pantry website](#). If you have further questions, please contact the Pantry Coordinator at usfpantry@usfca.edu or 415-422-4099 (during business hours Monday thru Friday from 9:00am - 5:00pm). You can find out about additional food security resources through the [USF food insecurity resource page](#) and the [CalFresh resources site](#).